Question 25

25 In order to keep the store properly stocked with produce, a grocery store manager wants to determine the most popular day of the week for people to buy apples and oranges. She randomly collects a sample of sales records for each day for three consecutive weeks. Is this type of data collection method a survey, an observational study, or an experiment? Explain your reasoning.

It's an observational study, because the manager isn't asking anyone about their preferences neither he is changing conditions of the experiment, so I can eliminate these two options. What I'm left with is an observational study.

Answer: observational study.

Question 26

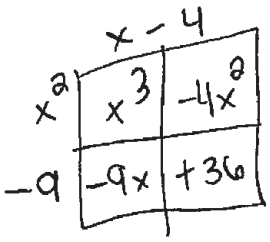
26 Given the following functions $f(x) = 2x^2 + 3x - 7$ and $g(x) = 2x - 5$, determine $f(x) - g(x)$, and express the result in standard form.

$$(2x^2+3x-7)-(2x-5)$$

$$2x^2+3x-7$$
$$-2x+5$$
$$2x^2+x-2$$

Question 27

27 Determine algebraically the zeros of $f(x) = x^3 - 4x^2 - 9x + 36$.



$f(x) = x^3 - 4x^2 - 9x + 36$

$(x-4)(x^2-9)$

$0 = (x-4)(x+3)(x-3)$

$x-4=0$ $\quad x+3=0$ $\quad x-3=0$

$+4\ +4$ $\quad -3\ -3$ $\quad +3\ +3$

$x=4$ $\quad x=-3$ $\quad x=3$

$f(4) = 4^3 - 4(4)^2 - 9(4) + 36 = 0$

$f(-3) = -3^3 - 4(-3)^2 - 9(-3) + 36 = 0$

$f(3) = 3^3 - 4(3)^2 - 9(3) + 36 = 0$

Question 29

29 Express $\left(-2 + \sqrt{-25}\right)^2$ in simplest $a + bi$ form. Justify your answer.



$(-2+5i)^2$

$(-2+5i)(-2+5i)$

$4 - 10i - 10i - 25$

$-20i - 21$

Question 30

30 The most common radioactive form of cesium is cesium-137. The half-life of cesium-137 is 30 years.

Suppose 2.7 micrograms of cesium-137 are found in a soil sample. Write a function for the amount of cesium-137 remaining, $c(t)$, in micrograms, in terms of the number of years, t, since the soil sample was taken.

$c(t) = 2.7(0.5)^{t/30}$

Question 31

31 Solve algebraically for x:

$$\sum_{a=0}^{4}(ax-3)=5$$

$a=0$
$a=1$
$a=2$
$a=3$
$a=4$
} 5

$S_n = \frac{n(a_1 + a_n)}{2}$

$0(x)-3=$
$1(x)-3=$
$2(x)-3=$
$3(x)-3=$
$4(x)-3=$
} 5

$5 = \frac{5(-3+4x-3)}{2}$

$10 = \frac{5}{2}(-3+4x-3)$

$2 = (-3+4x-3)$

$2 = (4x-6)$

$8 = 4x$

$x = 2$

$-3+(x)-3+2x-3+3x-3+4x-3 = 5$

$-15+10x=5$

$10x=20$

$x=2$

Question 32

32 Given the exponential function $y = 5.381e^{0.214x}$, rewrite the function in the form of $y = a(b)^x$, rounding values to the *nearest thousandth*.

$y = 5.381(1.239)^x$

Question 33

33 Determine the quotient and remainder of the expression below.

$$\frac{2x^4 - 27x^2 + 33x - 8}{x^2 + 3x - 5}$$

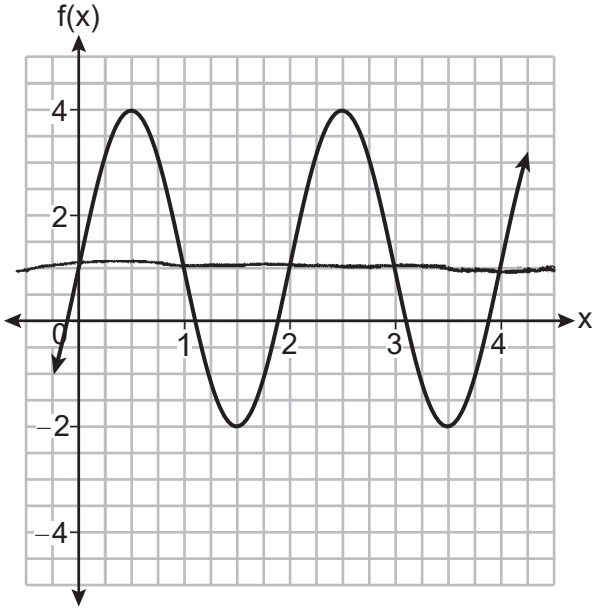
Express the result in the form $q(x) + \frac{r(x)}{d(x)}$.

$$\begin{array}{r} 2x^2 - 6x + 1 \\ x^2+3x-5 \overline{)\,2x^4 - 27x^2 + 33x - 8} \\ = 2x^4 - 6x^3 + 10x^2 \\ \hline -6x^3 - 17x^2 + 33x - 8 \\ +6x^3 + 18x^2 - 30x \\ \hline x^2 + 3x - 8 \\ -x^2 - 3x + 5 \\ \hline -3 \end{array}$$

$$\frac{2x^4-27x^2+33x-8}{x^2+3x-5} = 2x^2-6x+1+\left(\frac{-3}{x^2+3x-5}\right)$$

Question 34

34 The sinusoidal function f is graphed below.



State the period of the function f.

2

State the amplitude of the function f.

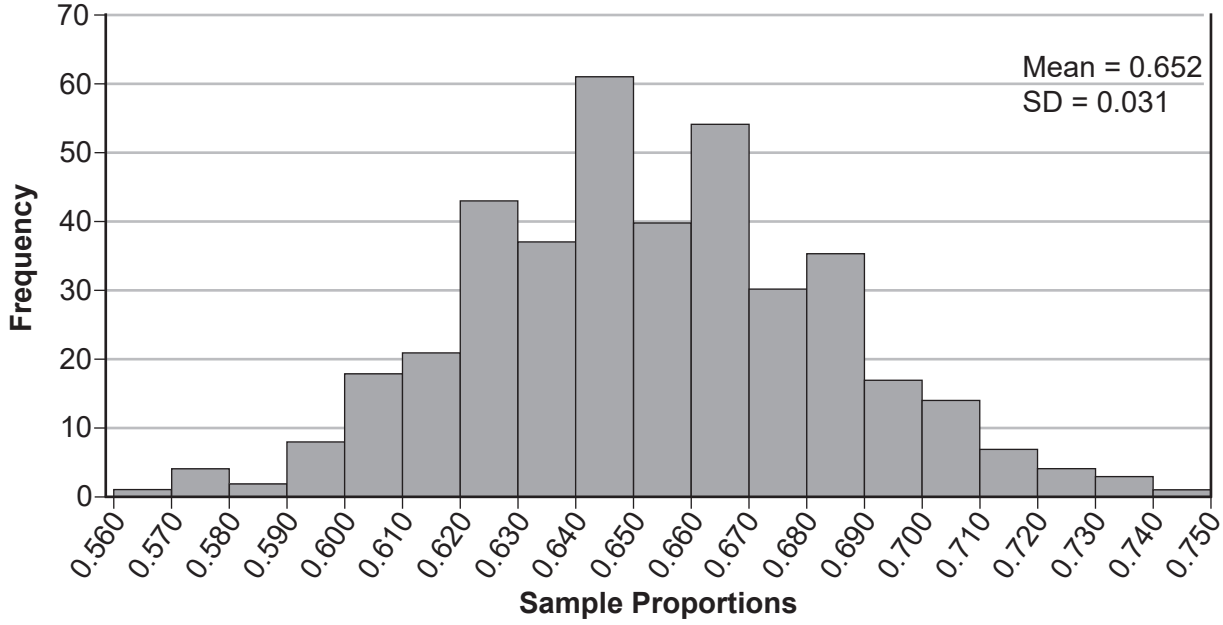
3

Write the equation in the form $f(x) = A \sin(Bx) + D$ to represent the sinusoidal function.

$f(x) = 3\sin(\pi x) + 1$

Question 35

35 A principal wonders what percentage of parents at her large high school restrict their child's cell phone privileges. The school surveys a random sample of 250 parents at the school and finds that 65% of them restrict their child's cell phone privileges. The school runs a simulation with 400 samples of 250 parents, assuming that 65% of them restrict cell phone privileges. The simulation results are shown below.



Use the simulation to determine an interval containing the middle 95% of simulated proportions. Justify your answer. Round all values to the *nearest hundredth*.

$0.652 \pm 2(0.031)$

$(.59, .714)$ $(.59, .71)$

The school district's superintendent worries that less than half of parents restrict their child's cell phone privileges. Do the simulation results support the superintendent's concern? Explain.

no because 50% falls below the 59%–71% interval.

Question 36

36 Lubin pours green tea into a cup to cool. The function below can be used to model the temperature, in degrees Fahrenheit, of Lubin's tea x minutes after he pours it into the cup.

$$T(x) = 110e^{-0.0453x} + 74$$

State the temperature of the tea, in degrees Fahrenheit, when Lubin poured it into the cup.

$110e^{-0.0453 \cdot 0} + 74$

$= 110 + 74 = 184°F$

To the *nearest tenth*, use $T(x)$ to determine the average rate of change in degrees per minute, from $x = 0$ to $x = 5$.

$$\frac{T(5) - T(0)}{5} = -4.5$$

Question 36 is continued on the next page.

Question 36 continued

Interpret the value of the average rate of change within the given context.

The temp decreases a rate of 4.5°F per minute on average.

To the *nearest tenth of a minute*, determine how long Lubin must wait after pouring his tea into the cup before the tea's temperature reaches 120° Fahrenheit.

$110e^{-0.0453x} + 74 = 120$

$110e^{-0.0453x} = 46$

$e^{-0.0453x} = 0.41818$

$-0.0453x = -0.8718$

$x = 19.24589$

$x = 19.2$